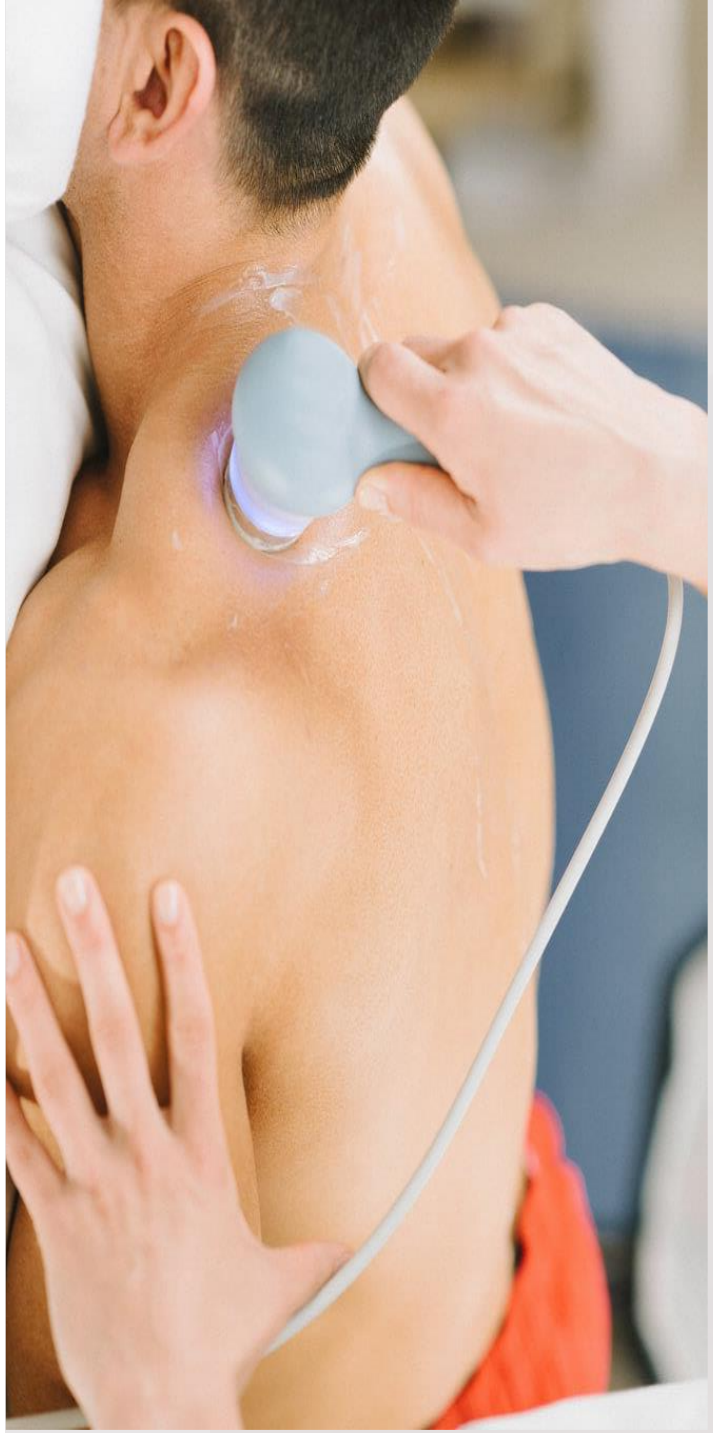
Electrotherapy III



Therapeutic Ultrasound

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Introduction



Fig.1 Ultrasound

Ultrasound (US) is a form of mechanical energy (not electrical), and therefore, **strictly speaking**, not really electrotherapy at all, but does fall into the Electro Physical Agents grouping. Mechanical vibration at increasing frequencies is known as Sound Energy.

Introduction



Fig.2 Ultrasound

Sound waves: Are longitudinal waves consisting, of a series of compression and rarefaction of the particles in the medium of transmission.

Sound waves from 20 Hz to 20,000 Hz are within the audible range. Beyond this upper limit, the mechanical vibration is known as Ultrasound.

Theory



Fig.3 Example of an ultrasound beam plot

Ultrasonic energy: Is a mechanical vibration identical to that of sound but of higher frequency. It is produced from electrical energy that is transformed into mechanical energy

Production of Ultrasound

The production of US is based on a phenomenon known as **piezoelectricity**

Piezo Generator
... solder an ultrabright LED
across the disks of a piezo...

Theory

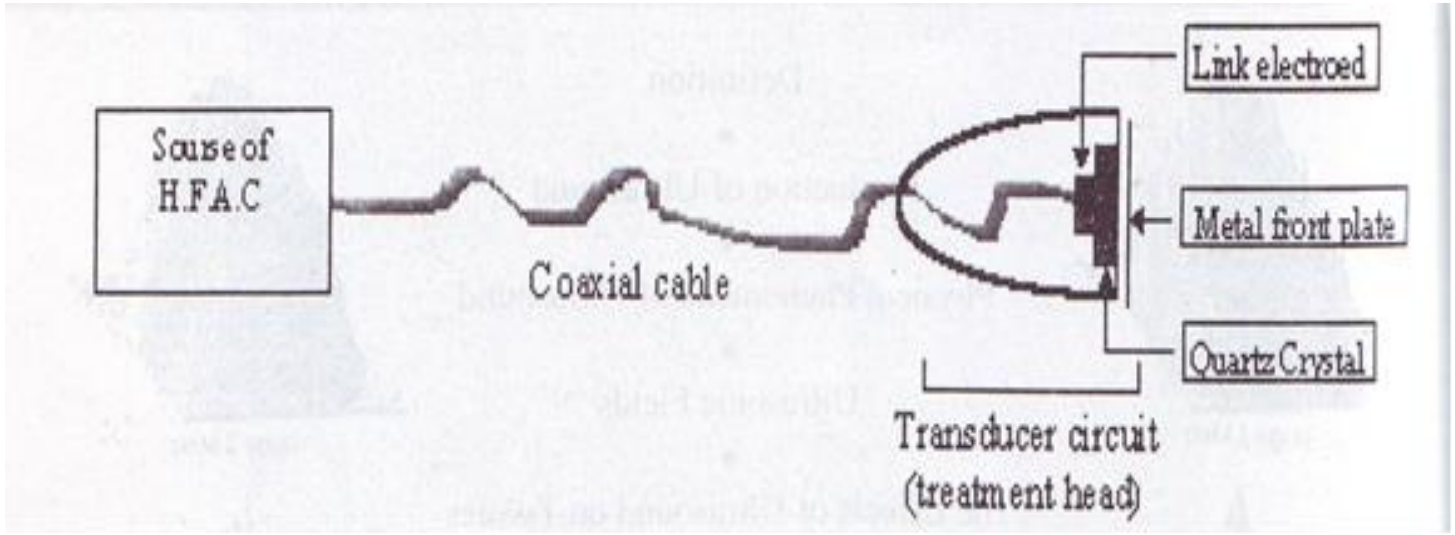
piezoelectricity



Fig.4 Example of an ultrasound DEVICE

is a natural phenomenon found in certain mineral crystals such as, germanium and quartz, but also be synthesized commercially. If this crystal compressed or deformed by mechanical means, it transforms the mechanical energy into electrical energy and vise versa.

(1) High frequency alternating current (H.F.A.C) converted by coaxial cable to transducer (treatment head).
(2) H.F.A.C. inside transducer applied to piezoelectric Crystal by link electrode. The crystal being fused to metal front plate.
(3) The crystal will contract and expand under the influence of an electrical current (deform).
(4) This change in the shape of crystal causes a movement of the metal front plate which produces an ultrasonic wave with specific frequency.

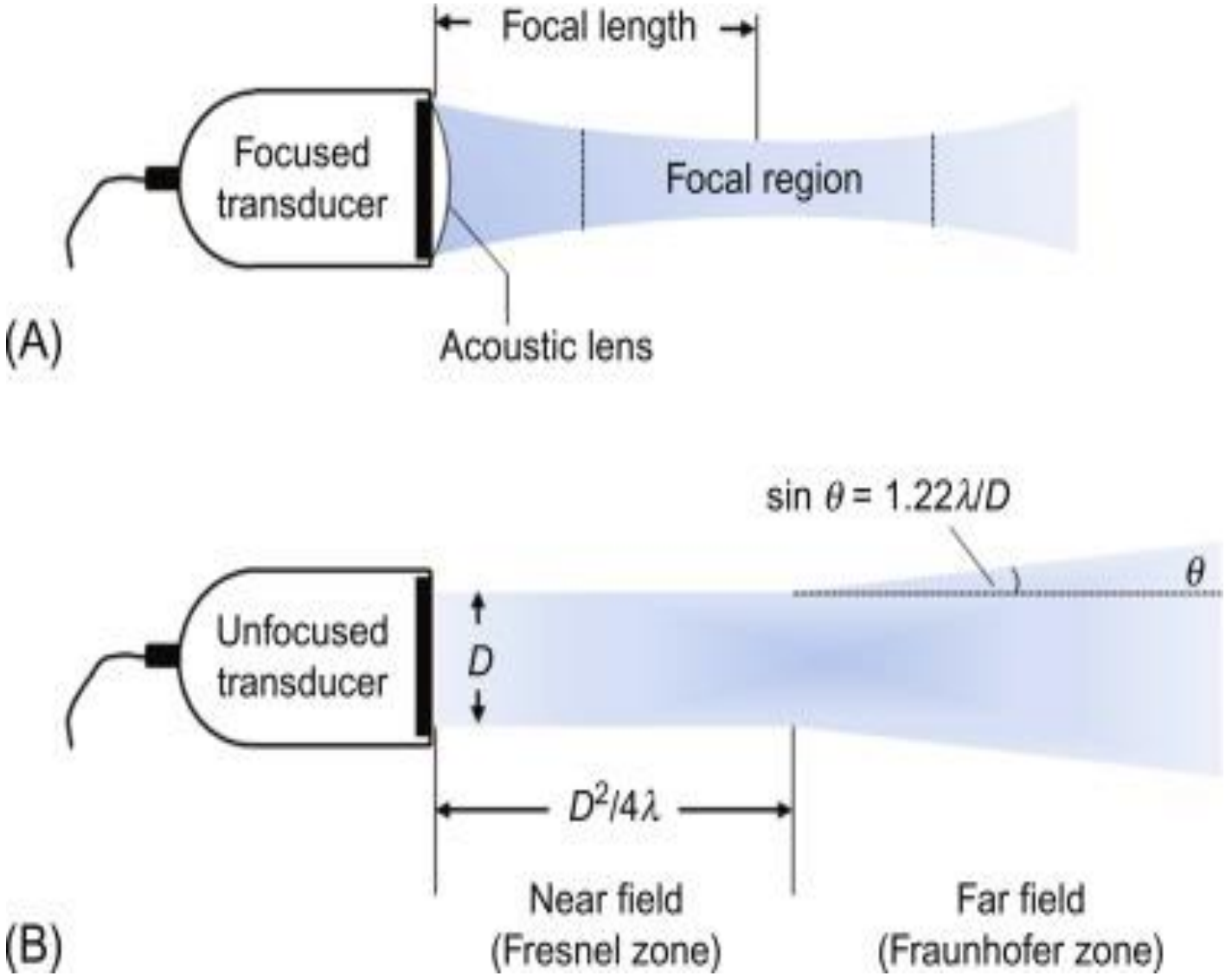


Ultrasound production

Theory

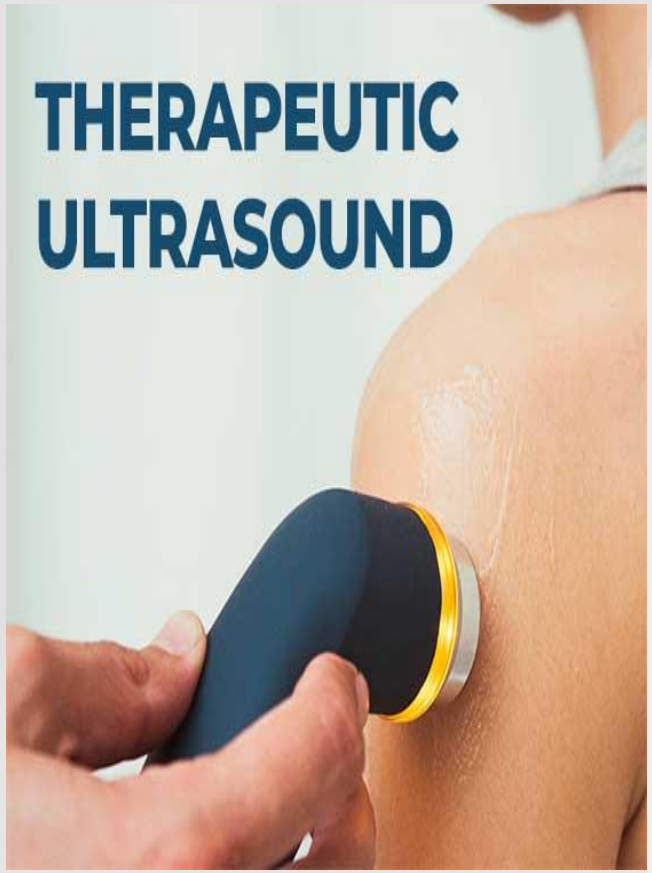


Fig.2 Example of an ultrasound beam plot



Therapeutic Ultrasound

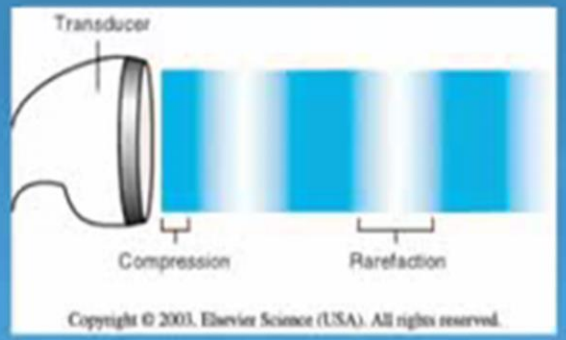
Theory



What is Ultrasound?

- High frequency sound
  - Defined as sound with freq > 20,000 Hz
  - Sound: Transmits energy by compressing and rarifying material
- Therapeutic US
  - 0.7-3.3 MHz





Therapeutic

The Physical Phenomena of Ultrasound

All materials (tissues) will present acoustic impedance to the passage of sound waves. The specific impedance of a tissue will be determined by its density and elasticity.

for the maximal transmission of energy from one medium to another, the impedance of the two media needs to be the same.

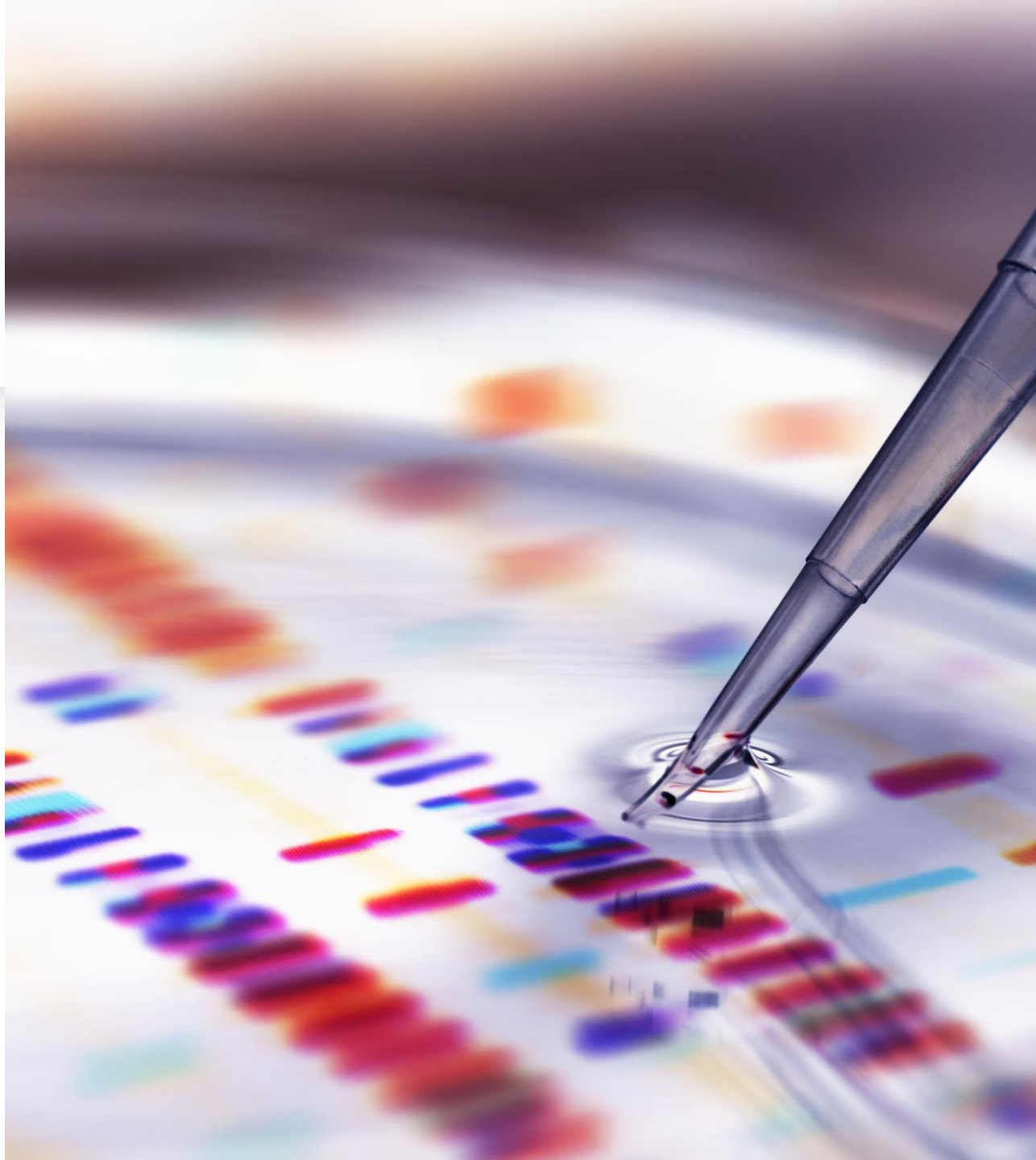
Clearly in the case of US passing from the generator to the tissues and then through the different tissue types, this can not actually be achieved



why

Because of;

1. ***Reflection***
2. **Coupling media**
3. ***Refraction***
4. **Absorption**
5. **Frequency of US**
6. **Attenuation**



Because of;

Reflection

Coupling media

Refraction

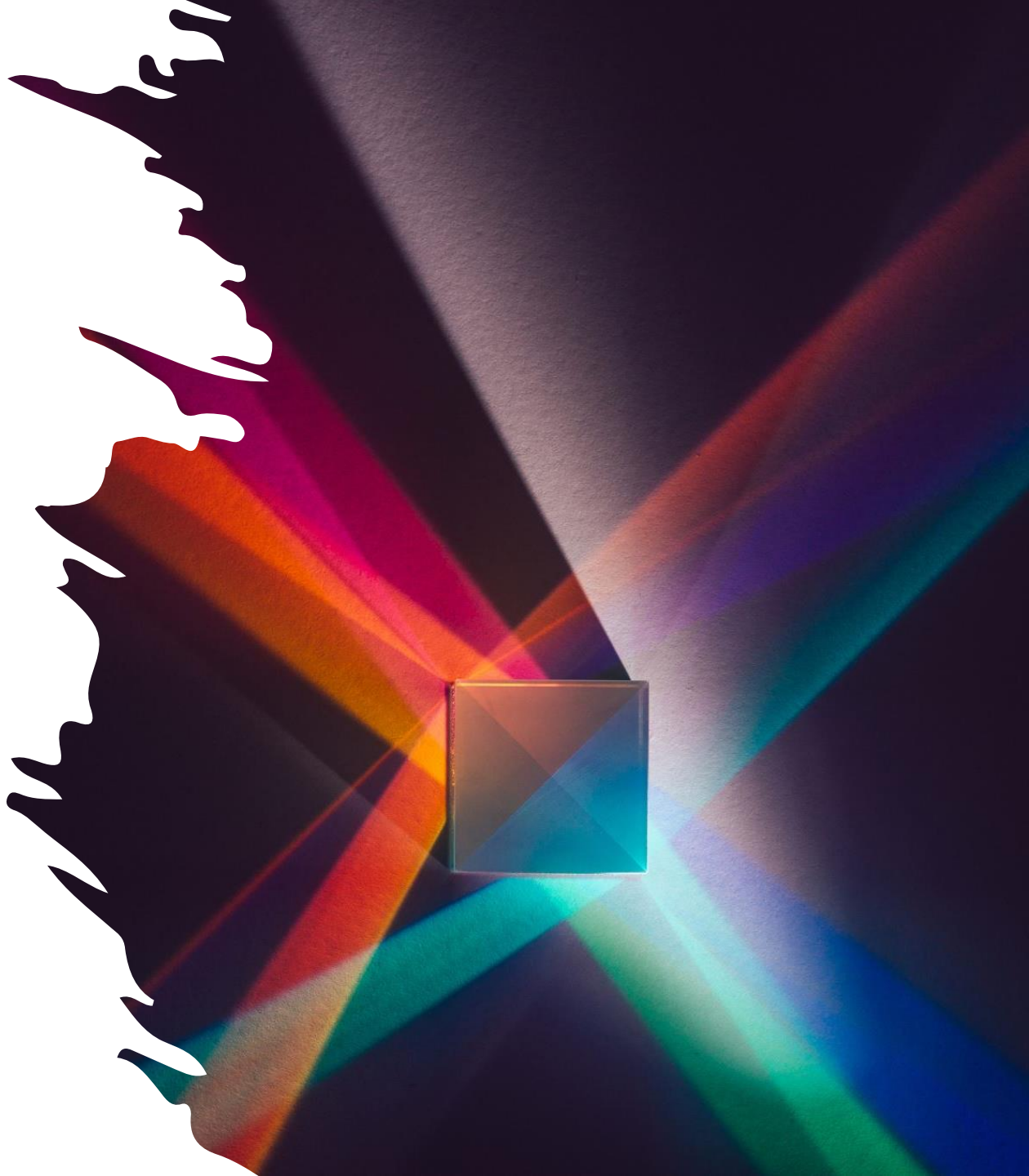
Absorption

Frequency of US

Attenuation

Reflection

- Reflection means the redirection of an incident beam away from a reflecting surface at an angle equal and opposite to the angle of incidence.

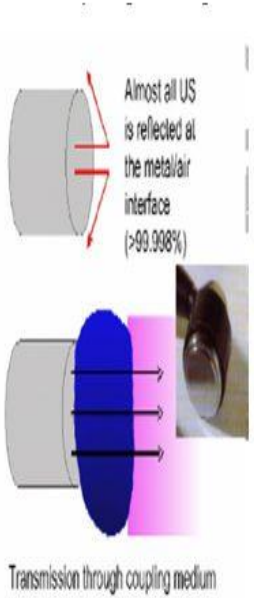


Reflection

- If there is an acoustic mismatch between the two media, reflection occurs.
- The amount of reflection depends on the difference in the acoustic impedance of the two media. The greater the difference in impedance at a boundary or interface, the greater the reflection that will occurs, and If there is an acoustic mismatch between the two media, reflection occurs. therefore, the smaller the amount of energy that will be transferred.

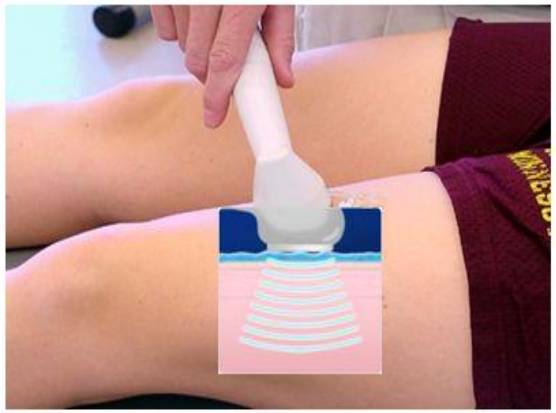
Impact of US Reflection in clinical setting

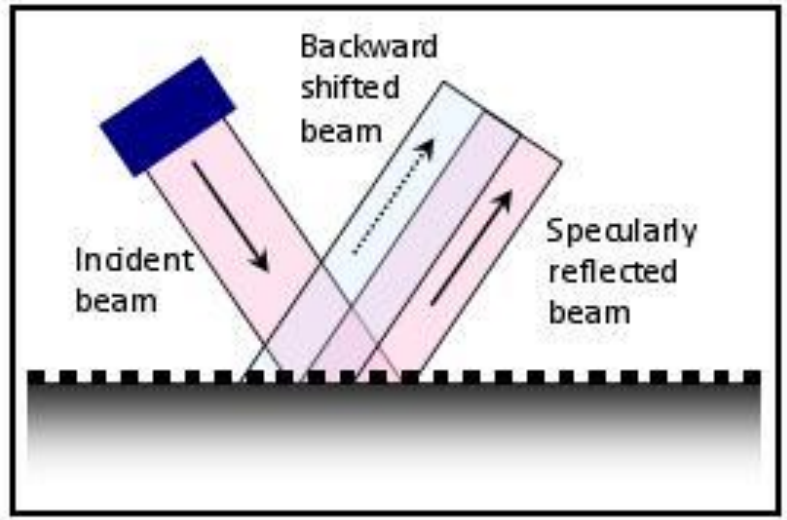
- Acoustic impedance
- Standing wave
  - Hotspots
  - Shearing forces

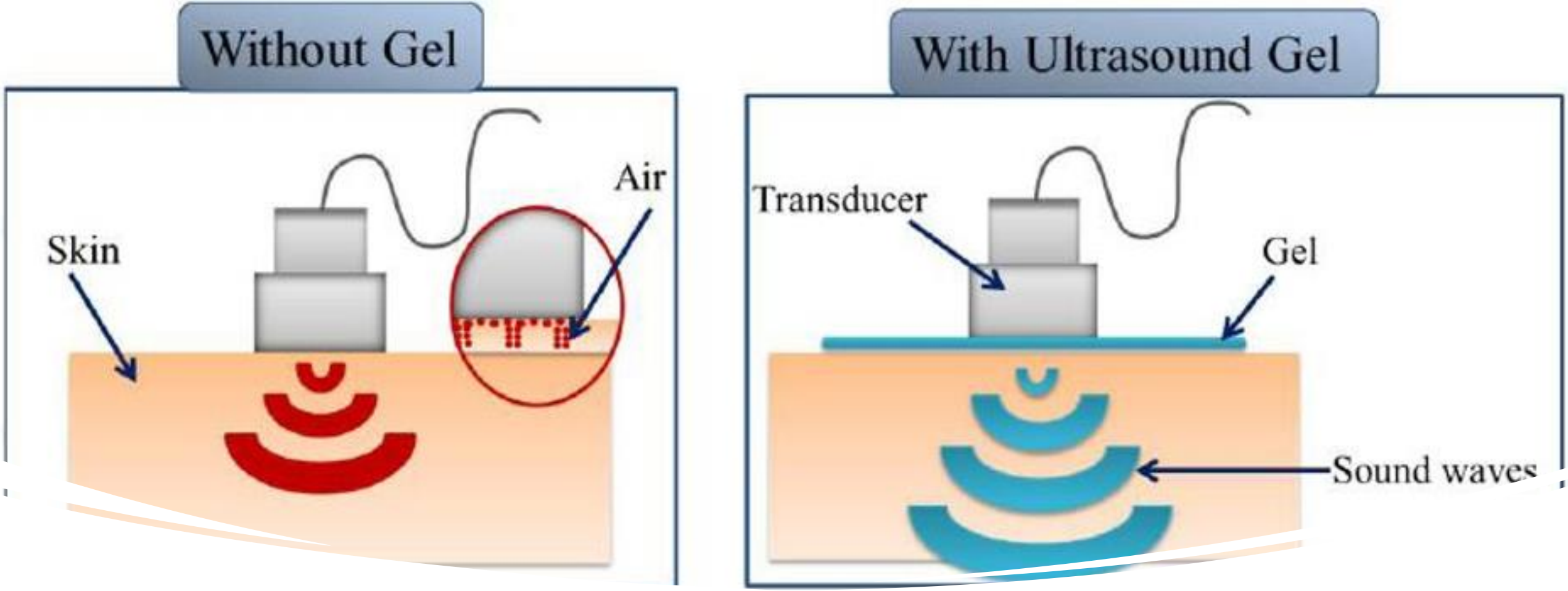


Clinical Application to Overcome Hot Spc

- keep the US transducer head perpendicular
- keep US in contact no air interface
- keeping US applicator in constant movement.
- Use of coupling media
- Using pulsed US







- The difference in impedance is greatest for the steel/air interface which is the first one which the US has to overcome in order to reach to body.
- In order to minimize this difference, a suitable coupling medium has to be utilized. If even a small air gap exists between the transducer and the skin the proportion of US which will be reflected approaches 99.998% which in effect means that there will be no transmission.

Acoustic impedance of different material

Material	Velocity	Density	Impedance
Air	340	0.625	213
Fatty tissue	1450	940	1.4×10^6
Water	1500	1000	1.5×10^6
Muscle	1550	1100	1.7×10^6
Bone	2800	1800	5.1×10^6
Steel	5850	8000	47×10^6

Percent reflection of ultrasound at different interface

INTERFACE	ENERGY REFLECTED (%)
Water-Soft tissue	0.2%
Soft tissue-Fat	1%
Soft tissue-Bone	15-40%
Soft tissue-Air	99.9%

Coupling media

The coupling media used in US transmission include water, various creams and gels. Ideally, the coupling medium should be

1- Fluid so as to fill all available spaces.

2- Relatively viscous so that it stays in place.

3- Have an impedance appropriate to the media it connects.

4- Allow transmission of US with minimal absorption, attenuation or disturbance.



Using water as coupling media



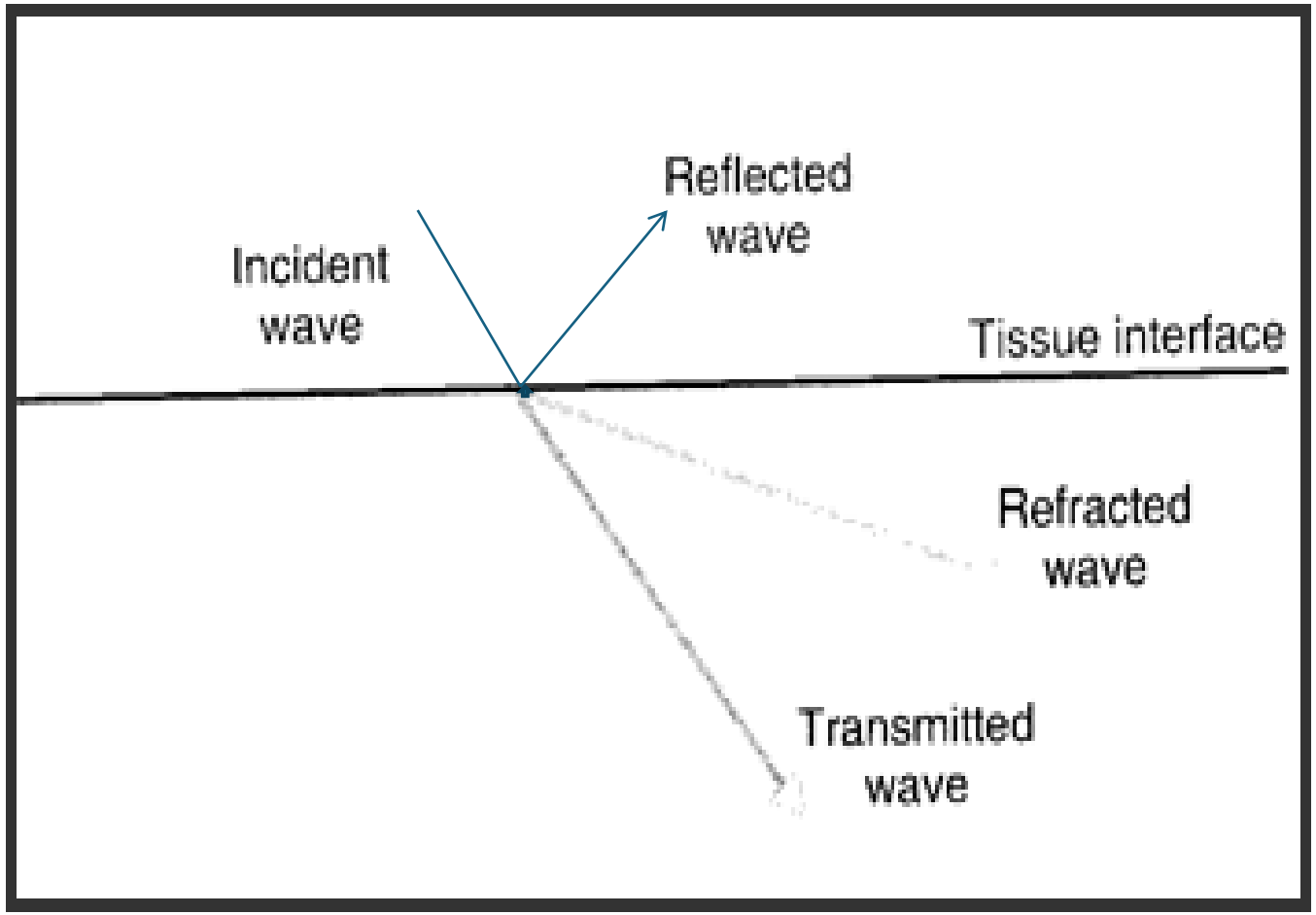
Figure 2



Refraction

- Refraction is the redirection of a wave at an interface surface. When refraction occurs, the ultrasound wave enters the tissues at one angle and continues through the tissue at different angle.
- If the US wave does not strike the boundary surface at 90°. Essentially, the direction of the US beam through the second medium will not be the same as its path through the original medium, its pathway is angled.

Refraction of ultrasound wave

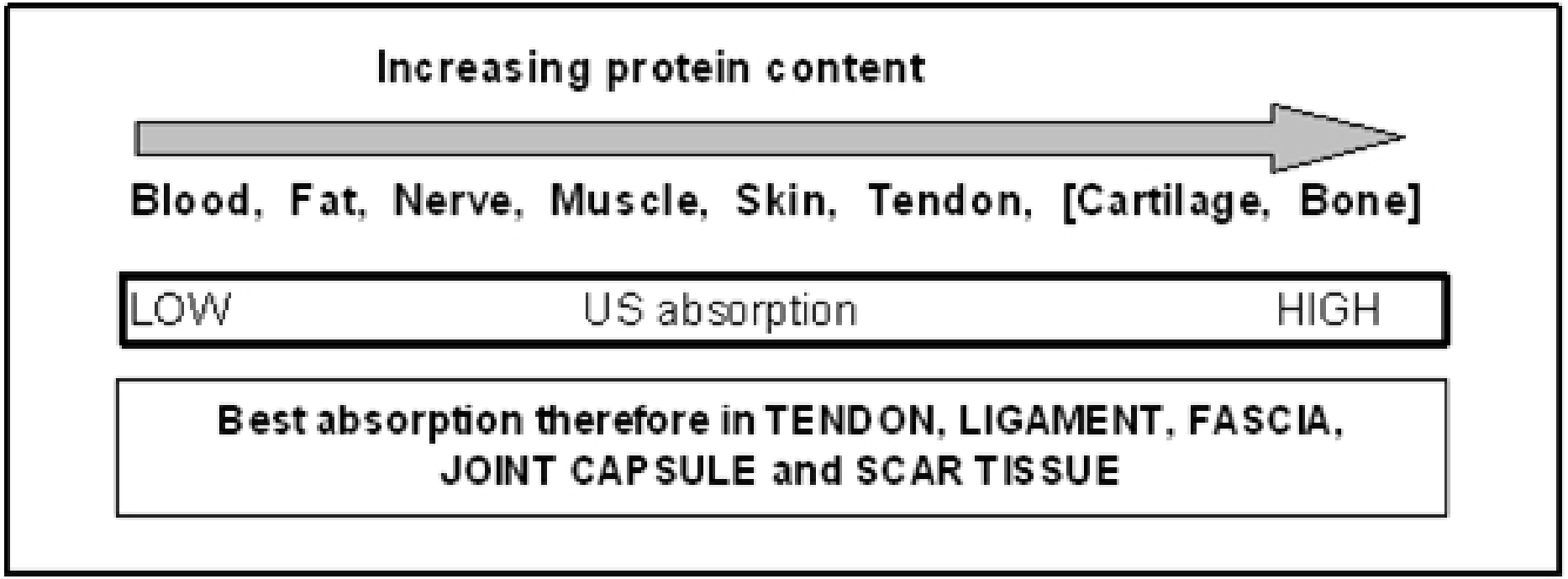


Absorption

- Absorption of US depends on nature of tissue and the frequency of the US. Some tissues are capable of greater absorption of US than others. Generally, the tissues with the **higher protein** content will absorb US to a **greater extent**.
- Tissues with high water content and **low protein** content will **absorb little** of the US energy (e.g. blood and fat) than tissues with a lower water content and a higher protein content will absorb US far more efficiently.



Absorption of US through different tissue



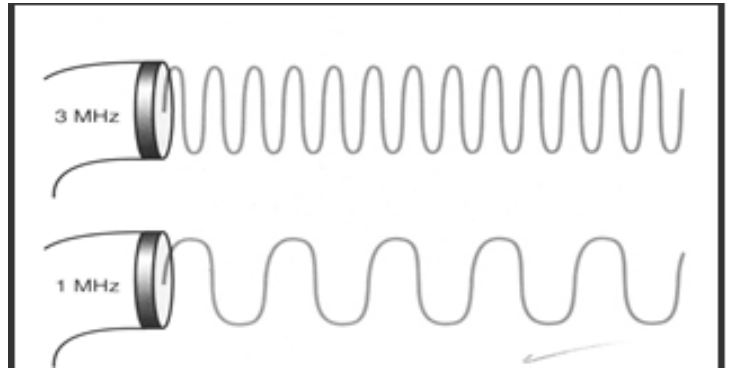
Frequency of US

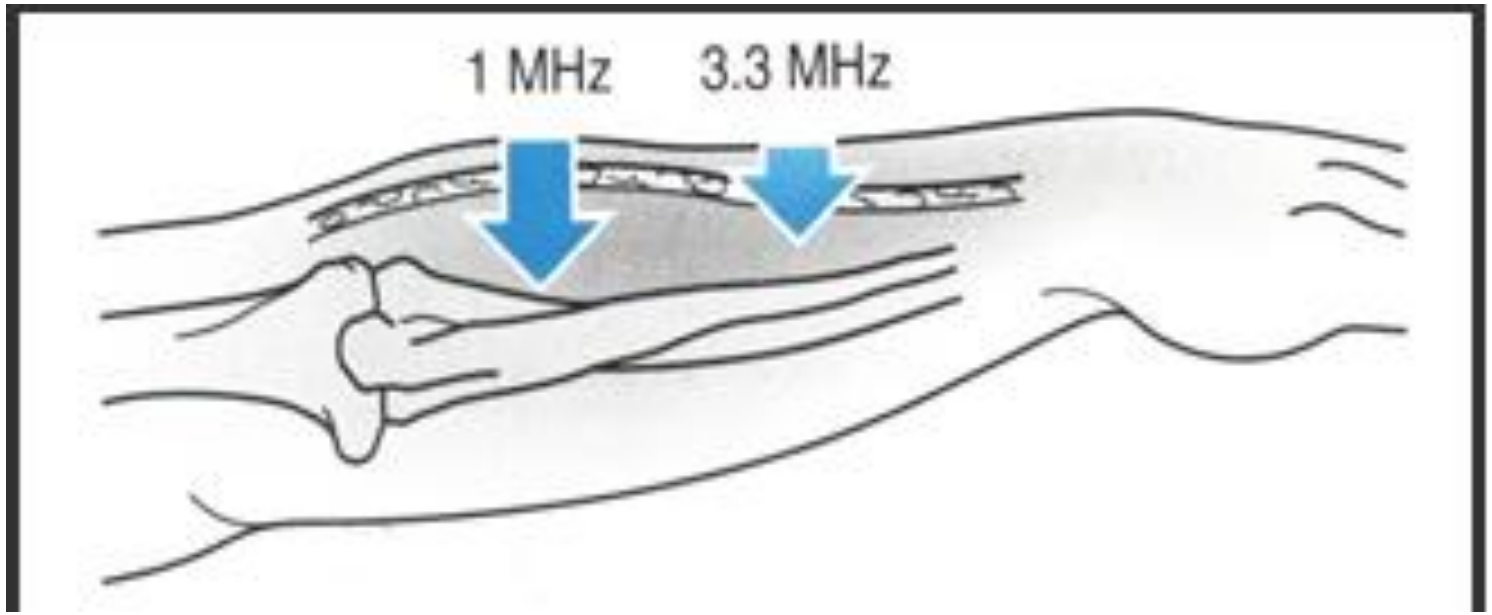
The number of compression-rarefaction cycles per unit of time, usually expressed in cycles per second (Hertz, Hz).

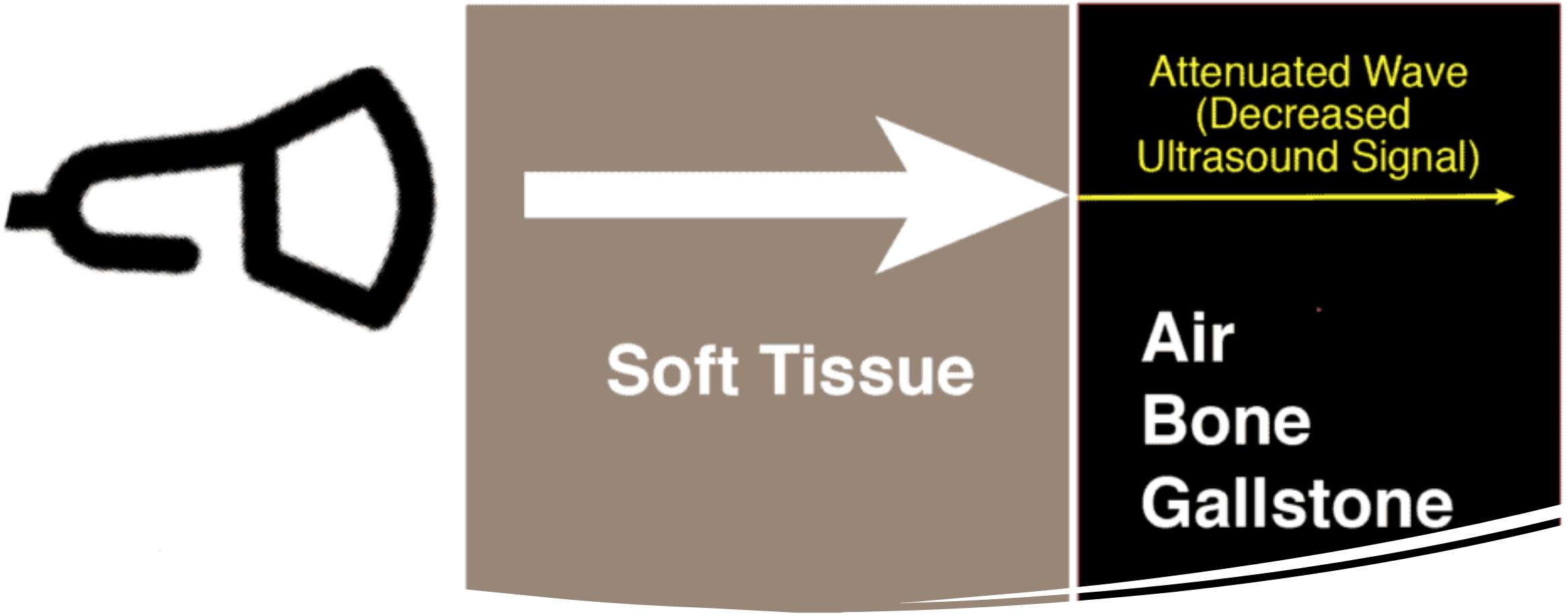
Increasing the frequency of ultrasound causes a decrease in its depth of penetration and concentration of the ultrasound energy in the superficial tissues

Ultrasonic frequency: 1 and 3 MHz









Attenuation

- Attenuation means loss of energy from the ultrasound beam in the tissue, and it depends on absorption and scattering. **Scattering** **is caused by reflection and refraction.**

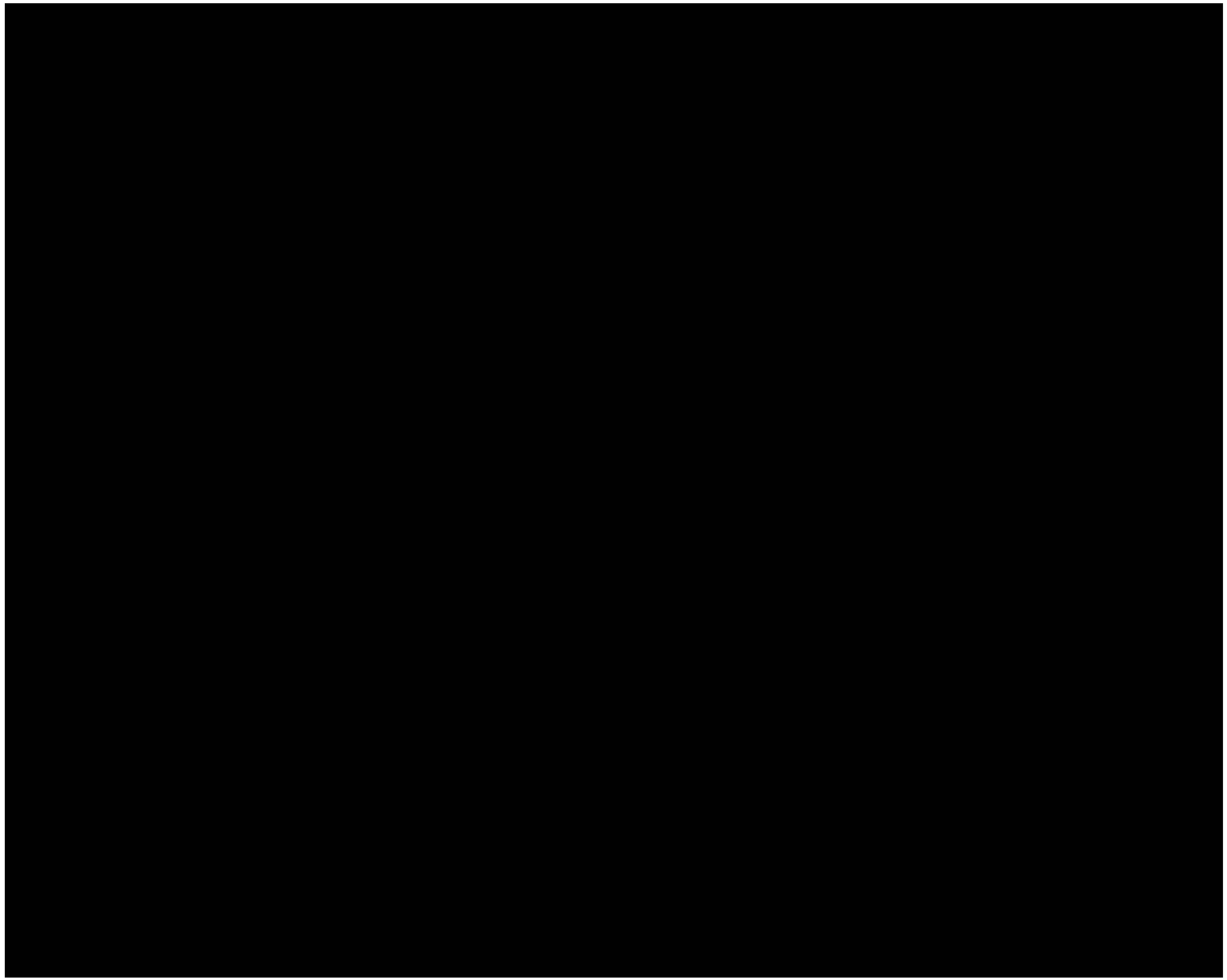
- There is an inverse relationship between the amount of energy that penetrate a material and the amount that is absorbed. Clearly, as the US beam penetrates into the tissues, a greater proportion of the energy will have been absorbed and therefore there is less energy penetrate

Physiological Effects of Ultrasound

- The physiological effects of US are generally divided into two main categories: **Thermal** and **Non- Thermal.**
- As the energy within the sound waves is passed and absorbed by the tissues, it will cause oscillation of the molecules of that tissues about their mean position, clearly any increase in the molecular vibration will result in heat generation.

Thermal effect of US

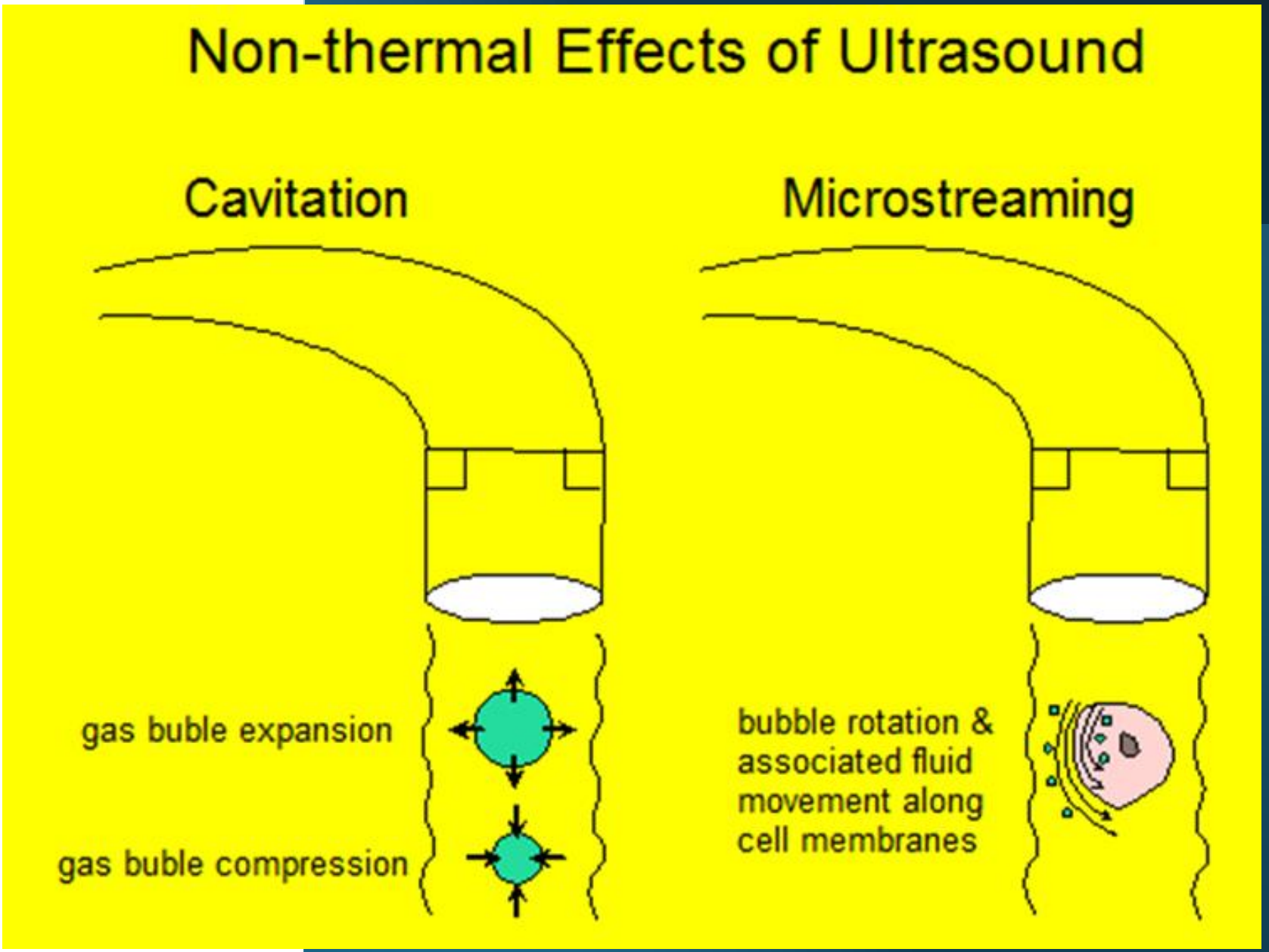
- The desirable effects of therapeutic heat can be produced by US can be used to selectively raise the temperature of particular tissues.
- Among the more effectively heated tissues are periosteum, superficial cortical bone, collagenous tissues (ligament, tendon and fascia) and fibrotic muscle. The greatest advantage is that deeply located area can be effectively heated as there is no loss of energy in the skin and subcutaneous fascia.



Non-Thermal Effects of Ultrasound

- The non-thermal effects of US fall into 3 main categories; Cavitation, Acoustic Streaming and Micro-Massage.
- **1- Cavitation** is the formation of gas bubble within the tissues and body fluids during the phase of relaxation or rarefaction of an ultrasound wave. There are 2 types of cavitation - **Stable and Unstable** which have very different effects.

Non-thermal Effects of Ultrasound



Cavitation

With stable cavitation, the bubbles oscillate in size throughout many cycles but do not burst. With unstable cavitation, the bubbles grow over a number of cycles and then suddenly implode.

This implosion produces large, brief, local pressure and temperature increases and causes free radical formation.

Cavitation

- **Stable cavitation**

Is not dangerous to the tissues as bubbles remain intact and oscillate harmlessly in the U.S field

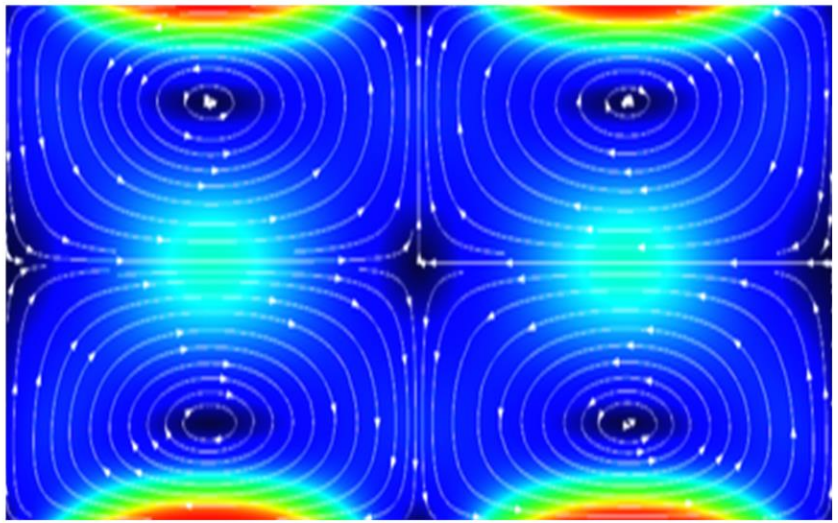
- **Transient cavitation**

Is dangerous to the tissues as the bubbles ground collapses rapidly in U.S causing increase in temperature.

The danger of cavitation can be minimized by:

1. **Using space averaged intensities below 4 watts/cm.**
2. **Using a pulsed U.S.**
3. **Moving the TREATMENT head during treatment.**

Acoustic Streaming



- is described as a small scale eddying of fluids near a vibrating structure such as cell membranes and the surface of stable cavitation gas bubble.
- This phenomenon is known to affect;
- Diffusion rates and membrane permeability, with a result that protein synthesis is enhanced.

Acoustic Streaming

- Sodium ion permeability is altered resulting in changes in the cell membrane potential.
- Calcium ion transport is modified which in turn leads to an alteration in the enzyme control mechanisms of various metabolic processes, especially concerning protein synthesis and cellular secretions.

stable cavitation and acoustic streaming

The result of the combined effects of stable ***cavitation and acoustic streaming*** *is that the permeability of the cell membrane altered which increase the activity levels of the whole cell.*

The US energy acts as a trigger for this process, but it is the increased cellular activity which is in effect responsible for the therapeutic benefits of the modality.

Micromassage

- **Micromassage;** is a mechanical effect in which the sound wave traveling through the medium will cause the molecules to vibrate. Possibly it is enhancing tissue **fluid interchange and affecting tissue mobility which is used to reduce edema.**

Does U.S may cause pain if it performed on bone?

YES

NO

Shearing effect

- **Shearing effect;** is the transformation of the longitudinal wave form into transverse wave form at the periosteum.
- As periosteum and bone tissue have different acoustic impedance, about 40% of the US energy is reflected along the periosteal surface (bone periosteum interface). This result is marked heating of the periosteum causing periosteal pain.

Therapeutic Effects of Ultrasound

Effect on blood flow

Continuous US can increase blood flow up to 45 minutes after treatment because of

1- Thermal effect.

2- Alteration of cell permeability

3- Histamine release

Therapeutic Effects of Ultrasound

- **Pain control**

Ultrasound can produce pain relief through direct and indirect mechanism

- Direct effect: US directly influence the cell membrane permeability of the nervous tissue resulting in;
- 1- Increase pain threshold
- 2- Decrease transmission of pain impulse along nerve fibers
- 3- produce counterirritant effect (gate theory for pain inhibition)

Therapeutic Effects of Ultrasound

- ***Indirect effect:*** result from the other effect of US which could lead to removing the source of pain as, increase capillary permeability and removing waste products and edema reduction, improving tissues healing, decrease muscle spasm and breaking down adhesion.

Therapeutic Effects of Ultrasound

- **Effect on Adhesion and scar tissues**
- Collagen has high absorption ability and is better heated with US. The thermal effect of US lead to increasing extensibility of collagen rich tissues, thus US could be used prior to range of motion exercises and stretching exercises.
- Also US are used to help improving the quality of the scar, resulting in a slightly strong and more elastic scar.

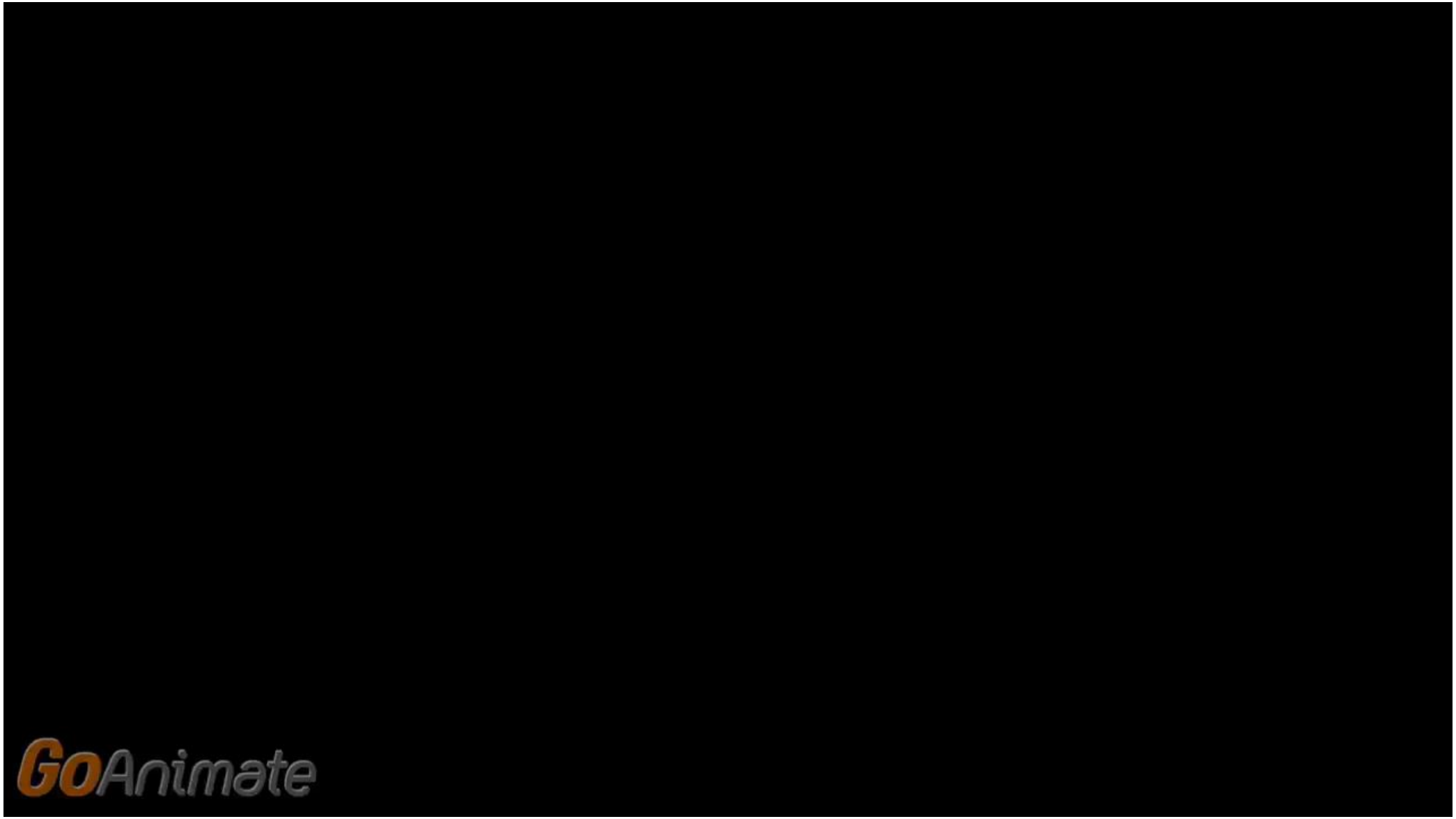

Why?

- Ultrasonic helps in breakdown adhesions by depolymersation of mucopolysaccharides, mucoproteins or glycoprotens or by reducing the viscosity of hyaluronice acid in joints.
- Adhesion and intercellular substances converted from the Gel to the Solid state responding to U.S. So resorption of adhesion occurs by both heating effect and micro massage effect of US.

Effect on wound healing

- Pulsed ultrasound at low intensities (0.8 $W.cm^2$) was effectively used to enhance healing of wound by:

1. stimulating collagen production by fibroblasts,
2. increases the intracellular calcium ion levels,
3. improve the local blood supply and encourage the growth of new capillaries,
4. in addition it stimulates the myofiboblasts to contract, giving rapid initial wound strength.


GoAnimate

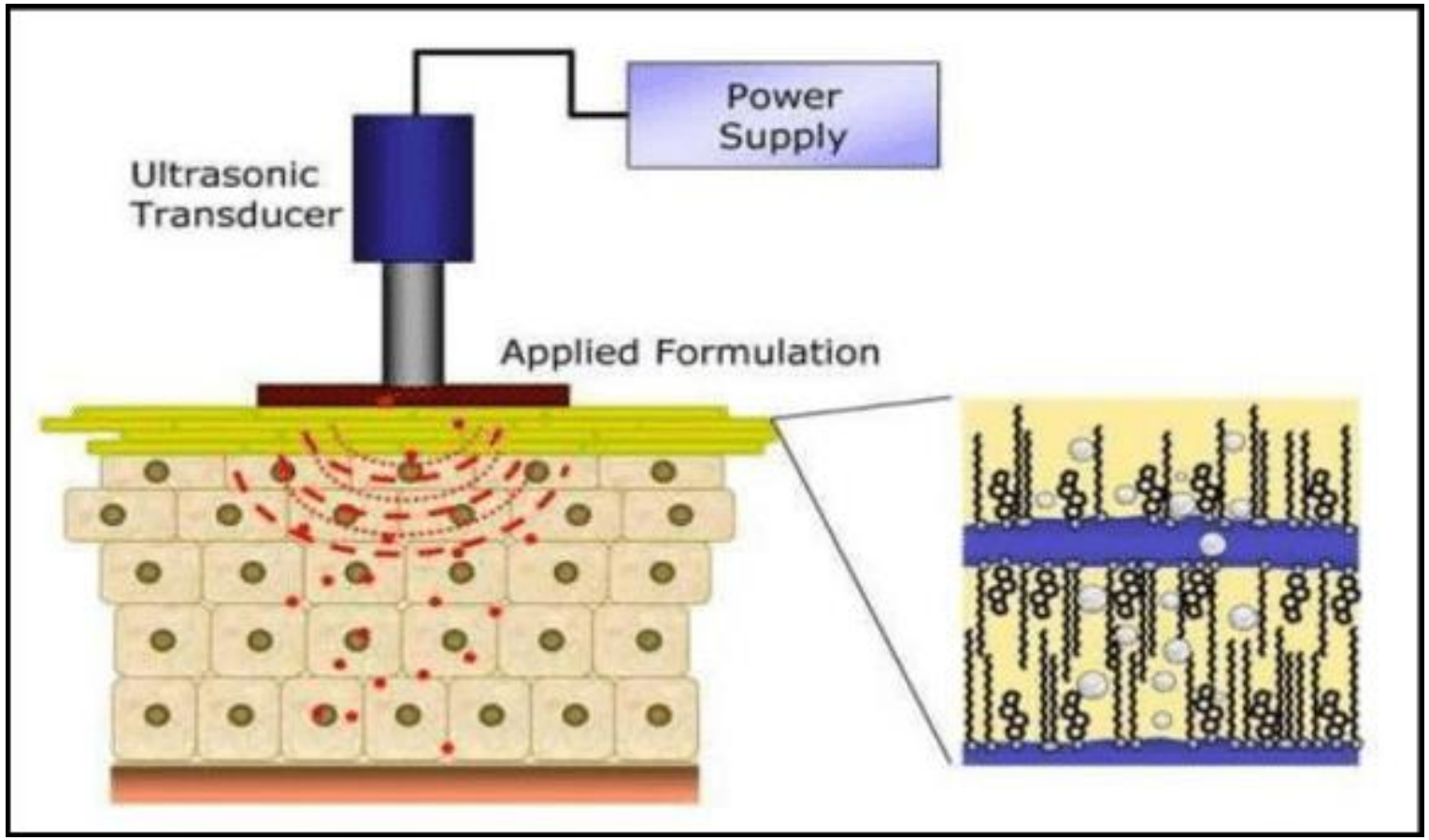
Fracture healing

- The use of ultrasound could accelerate the rate of fracture repair by a factor of 1:6. The US delivered a low intensity (0.03 W cm^{-2}) at 1.5 MHz pulsed at a ratio of 1:4. for 20 minutes. Using higher ultrasound doses (1- 2 W cm^{-2}) could have an adverse effect on the fracture healing process.

Phonophoresis

- It is the introduction of the molecules of drugs through the intact skin by the mechanical effect of ultrasound.
- It depends on the molecular weight of the drug, skin pores, frequency, intensity of insonation.
- Examples (Hydrocortisone, Nonsteriodal anti inflammatory drugs, Lidcocain, Salicylate)

phonophoresis



Modes of ultrasound

- **Continuous Ultrasound**

Continuous US causes primary thermal effects and capable of heating tissues located at up to 6 cm deep, according to the frequency used. Usually used in treatment of chronic lesions

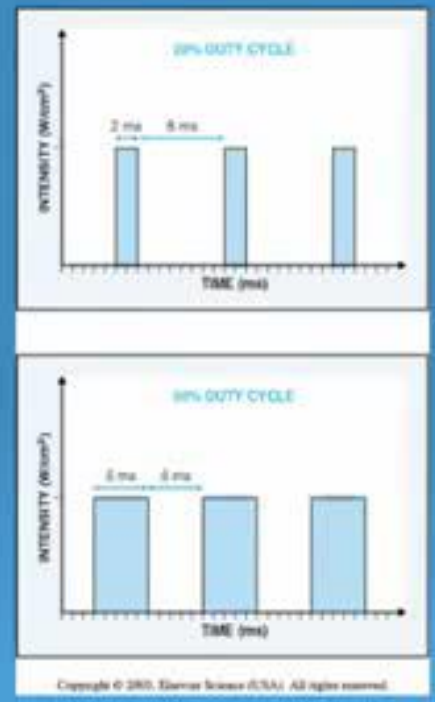
- **Pulsed Ultrasound**

Pulsing the ultrasound beam decrease the average intensity of the output and reducing the thermal effects and allowing for the non-thermal effects.

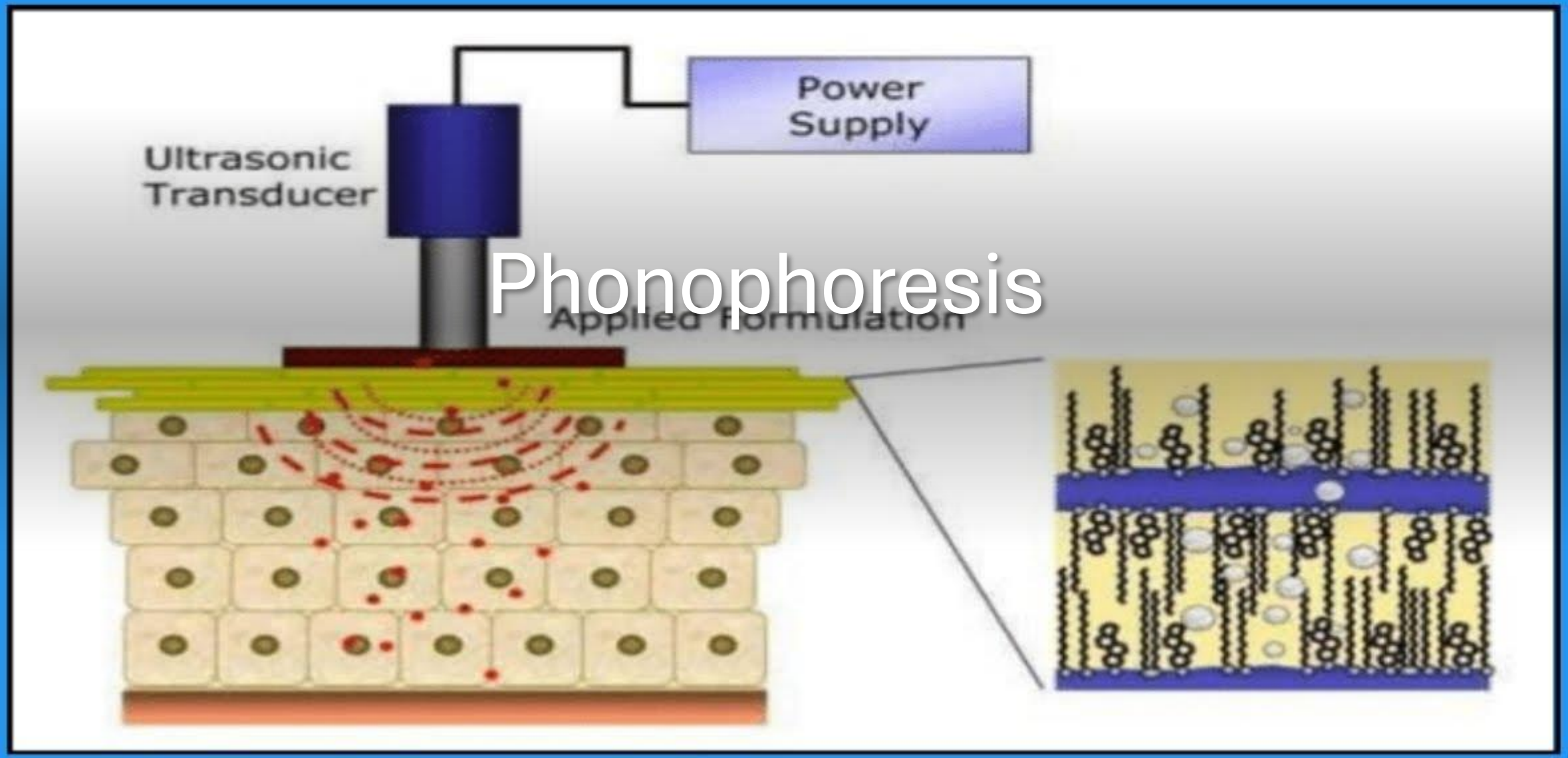
Duty Cycle

Duty Cycle

- The portion of the total time that the US is on to the total cycle time
- Example:
  - 5 ms on, 5 ms off
  - 5 on/10 total *100 = 50%
  - 2 on/10 total *100 = 20%



Phonophoresis

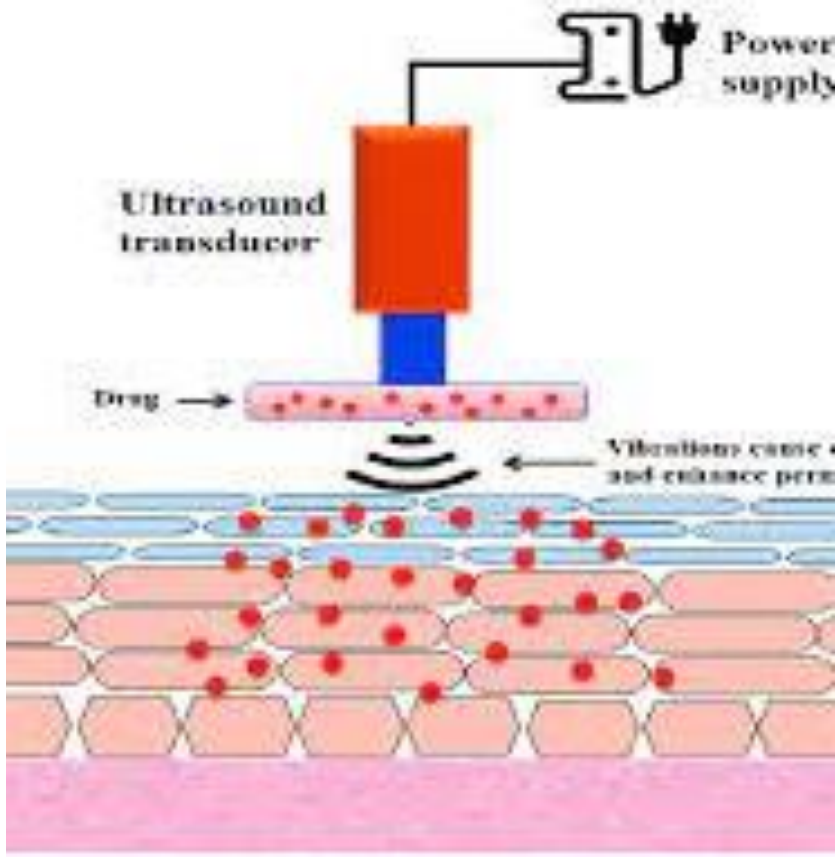


Phonophoresis

- Phonophoresis treatment is widely used in the field of physiotherapy, where the topically applied gel or cream-based medicines are being used along with ultrasonic therapy to accelerate the effects of ultrasound .
- Phonophoresis, also known as sonophoresis, has been claimed to intensify the percutaneous absorption of certain pharmacological agents such as anti-inflammatory steroids and local anesthetics from intact skin into the underlying subcutaneous structures by ultrasound, thereby improving their effectiveness.

Phonophoresis

- This procedure is commonly used in physical therapy practices. The procedure generally utilizes an ultrasound apparatus that produces frequencies of 0.7 to 1.1 MHz.
- The ultrasound intensities that are being employed usually range from 0.0 to 3.0 Watts per cm2. Both continuous-mode as well as pulse mode applications are utilized, and most treatments last from 5 to 8 mins, with the exception of treatments of larger areas (greater than 36 cm2) requiring more than 8 mins



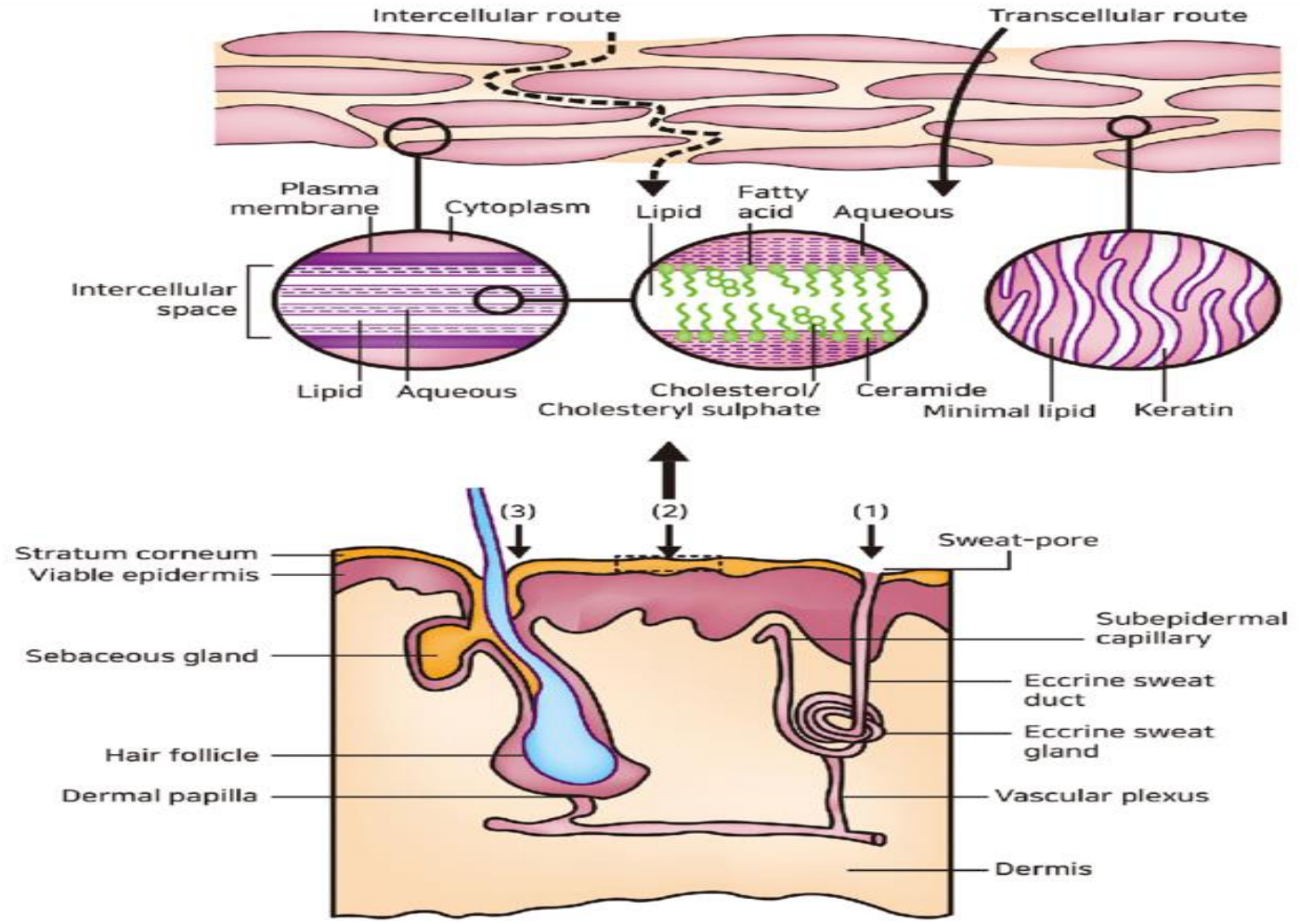
The therapeutic benefits of phonophoresis

- the therapeutic benefits of phonophoresis (such as pain relief and improved range of motion) when it was used to treat lateral epicondylitis, temporomandibular joint pain, and osteoarthritic conditions.
- Recent studies have shown that, when compared with placebo treatments or ultrasound alone, phonophoresis provides clinical improvement by decreasing pain and increasing function .Phonophoresis has been suggested by early studies to enhance the absorption of analgesics and anti-inflammatory agents.



Mechanism of action in phonophoresis

- Continuous ultrasound has thermal effects, and pulsed ultrasound has mechanical effects such as cavitation, acoustic streaming, microstreaming, increased skin pore size, and increased pore numbers and intercellular space.
- Therefore, the tissues undergo several changes via their interaction with therapeutic ultrasound waves. The general result is that skin permeability is enhanced by the augmented mechanical stress and/or by creation of permanent or temporary cavities through corneocytes and keratinocytes



Route (1) : Sweat duct
(2) : Across the continuous stratum corneum (Transepidermal transport)
(3) : Hair follicle with their associated sebaceous glands

- there are various medicinal gels that are used for the trans dermal drug delivery by the phonophoresis. The effectiveness and penetrating power vary according to the properties of the gel. **Cardero et al.** indicated that diclofenac has the highest transdermal penetration among NSAIDs such as indomethacin, piroxicam, tenoxicam, ketorolac and aceclofenac.
- Rosim et al., also recommended that therapeutic ultrasound administration increases the percutaneous absorption of the topical **diclofenac gel**. According to this, diclofenac is considered to be a good agent for phonophoresis administration

Gene therapy

- Another future application for ultrasound as a topical enhancer that seems to show promise lies in the field of topical gene therapy. There is considerable interest in facilitating the transfer of genes into diseased tissues and organs.
- The main aim is to increase the delivery efficiency of exogenous nucleic acid to the intended target. The ideal system would enhance gene expression in the target while having no effect in non-target tissues.
- Ultrasound might be able to provide this localization. Ultrasound has been shown to enhance gene transfer into cells in vitro and in vivo.
- Significantly better transfection is achieved in the presence of cavitation. Enhanced gene transfer is found either when the exposed bubbles are in the vicinity of the genetic material or when genes are encapsulated within or bound to the bubbles.